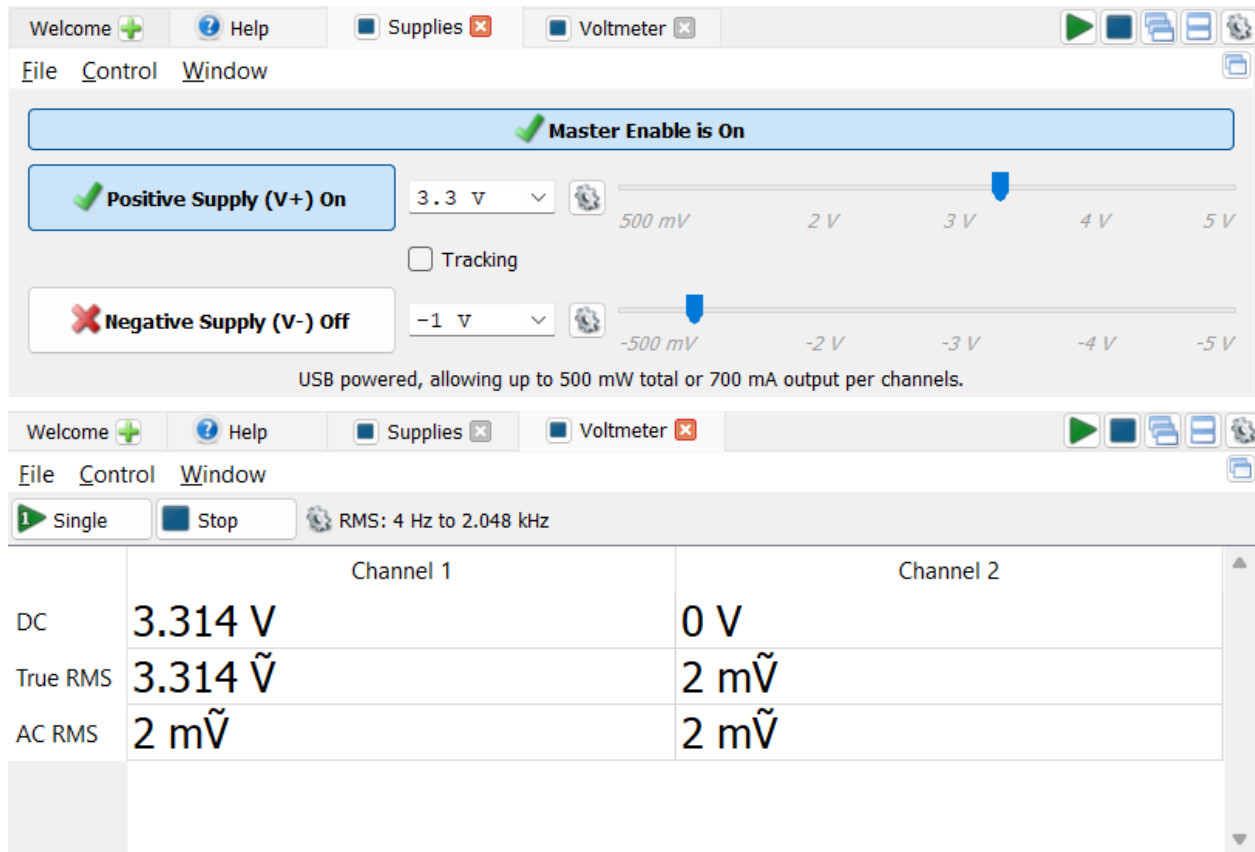
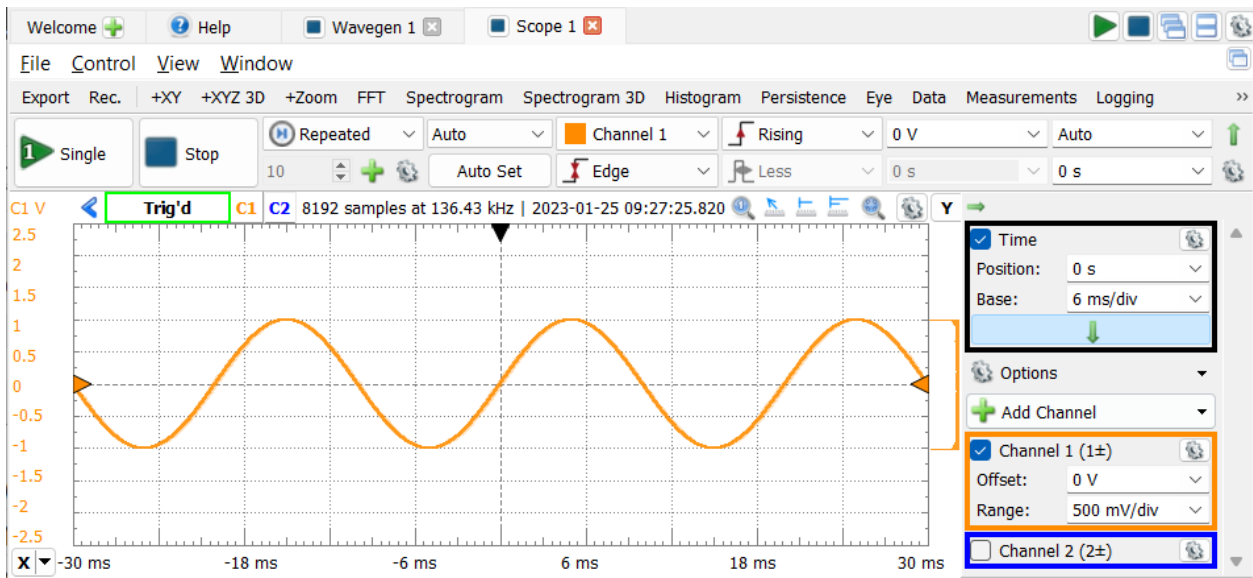
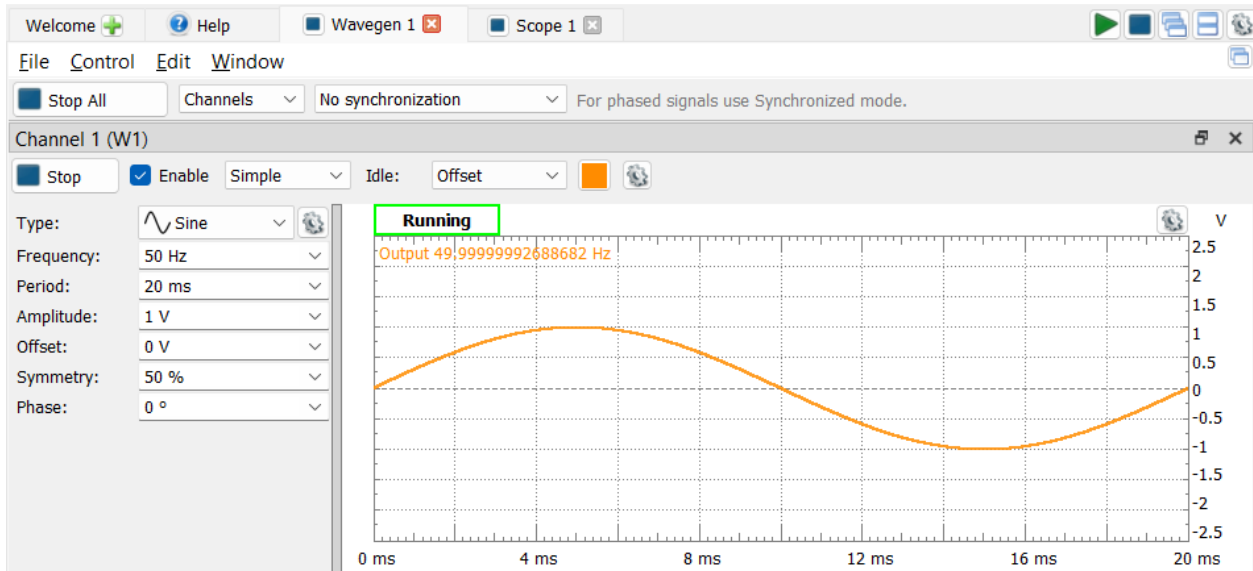


rHayden Fuller
Generate and measure directly:
DC:



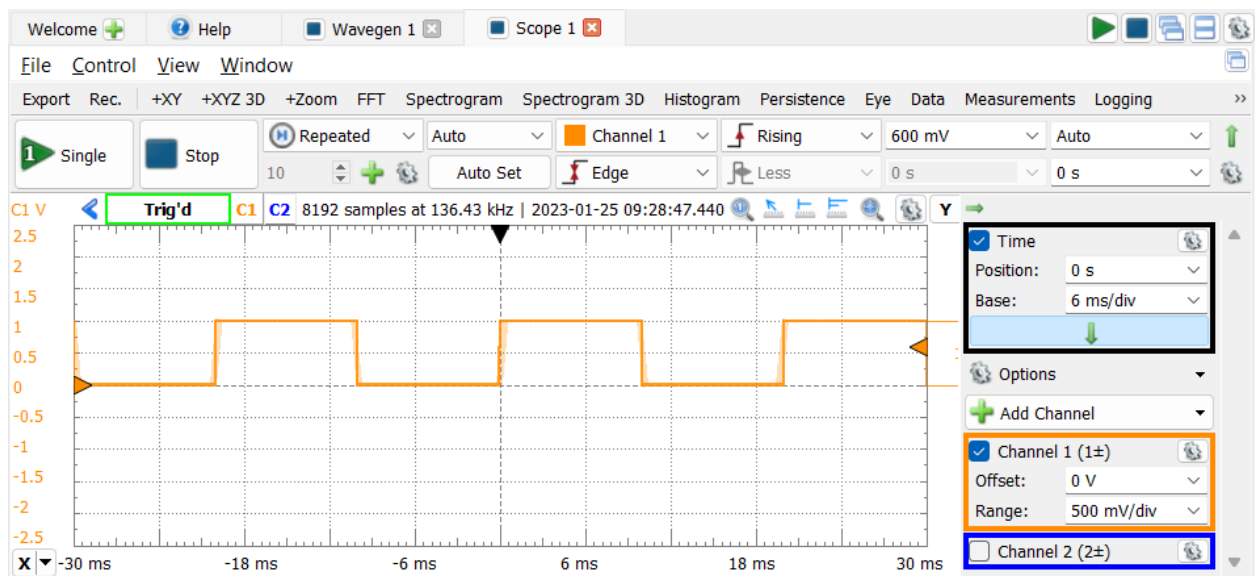
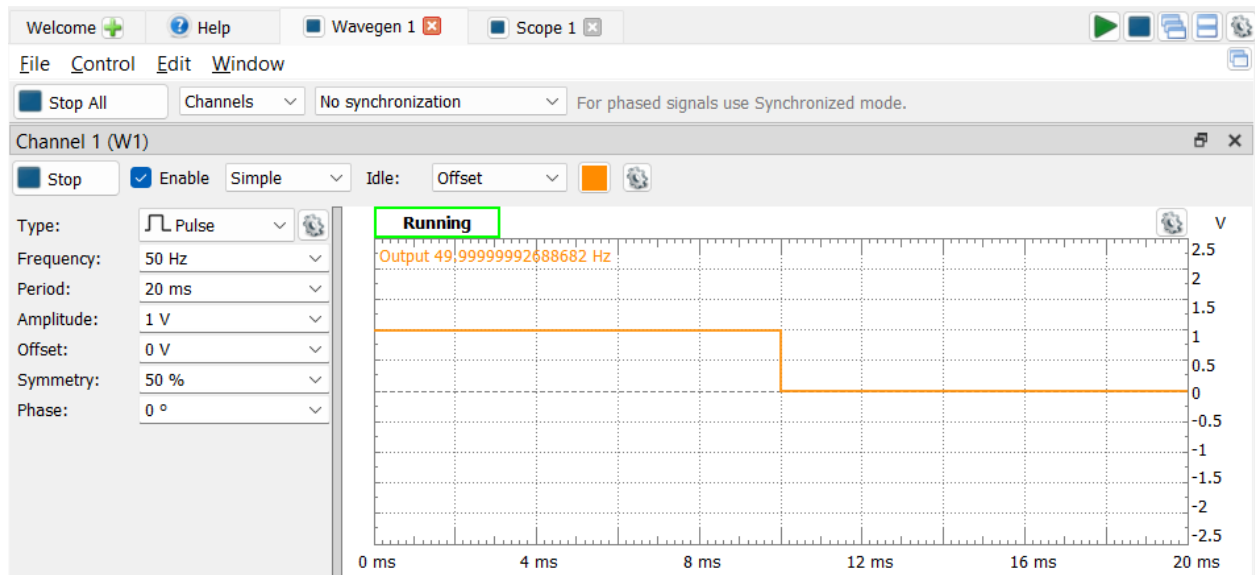
Sinusoid:

W1 ————— **1+**

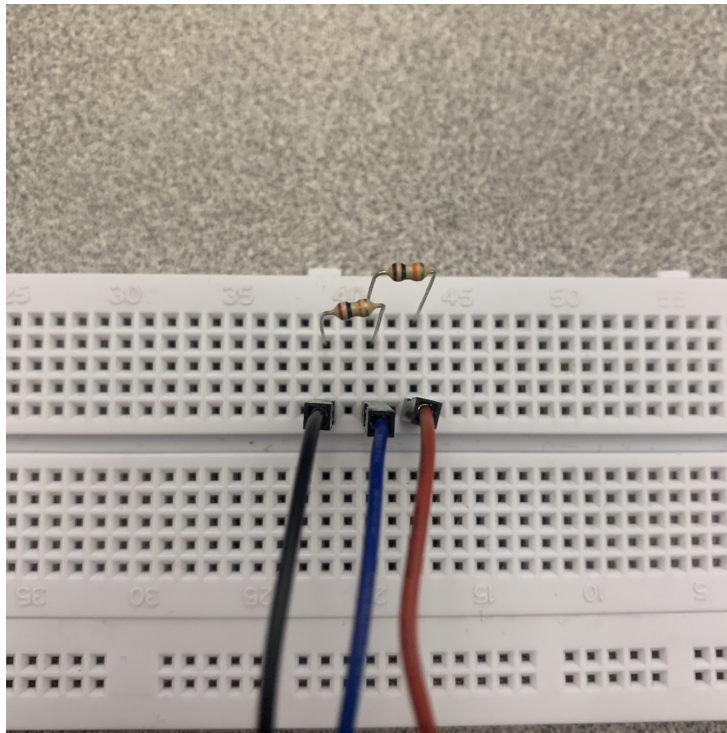
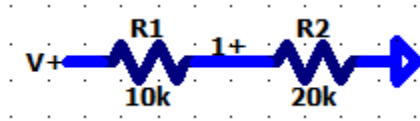


Pulsed signal:

W1 ————— **1+**



Measure DC across one resistor: Simple 10k and 20k in series (20k on gnd side which will be measured across for the remaining objectives)

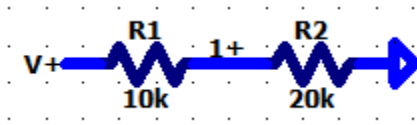


5v:

Red: V+, 5V

Black: Gnd

Blue: 1+, Channel 1



Software interface showing supply settings. The top bar includes 'Welcome', 'Help', 'Supplies', and 'Voltmeter'. The main area displays 'Master Enable is On' with a green checkmark. Below this, the 'Positive Supply (V+) On' section shows a setpoint of 5 V and a range from 500 mV to 5 V. The 'Negative Supply (V-) Off' section shows a setpoint of -5 V and a range from -500 mV to -5 V. A 'Tracking' checkbox is present. A note states: 'USB powered, allowing up to 500 mW total or 700 mA output per channels.' A 'System Monitor' checkbox is at the bottom.

Software interface showing measurement results. The top bar includes 'Welcome', 'Help', 'Supplies', and 'Voltmeter'. The main area displays 'Single' and 'Stop' buttons, and a frequency range 'RMS: 4 Hz to 2.048 kHz'. The results table is as follows:

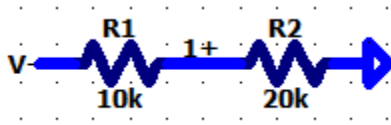
	Channel 1	Channel 2
DC	3.33 V	0 V
True RMS	3.33 V	2 mV
AC RMS	2 mV	2 mV

-5V:

Red: V-, -5V

Black: Gnd

Blue: 1+, Channel 1



Welcome Help Supplies Voltmeter

File Control Window

Master Enable is On

Positive Supply (V+) Off 5 V 500 mV 2 V 3 V 4 V 5 V

☐ Tracking

Negative Supply (V-) On -5 V -500 mV -2 V -3 V -4 V -5 V

USB powered, allowing up to 500 mW total or 700 mA output per channels.

☐ System Monitor

Welcome Help Supplies Voltmeter

File Control Window

Single Stop RMS: 4 Hz to 2.048 kHz

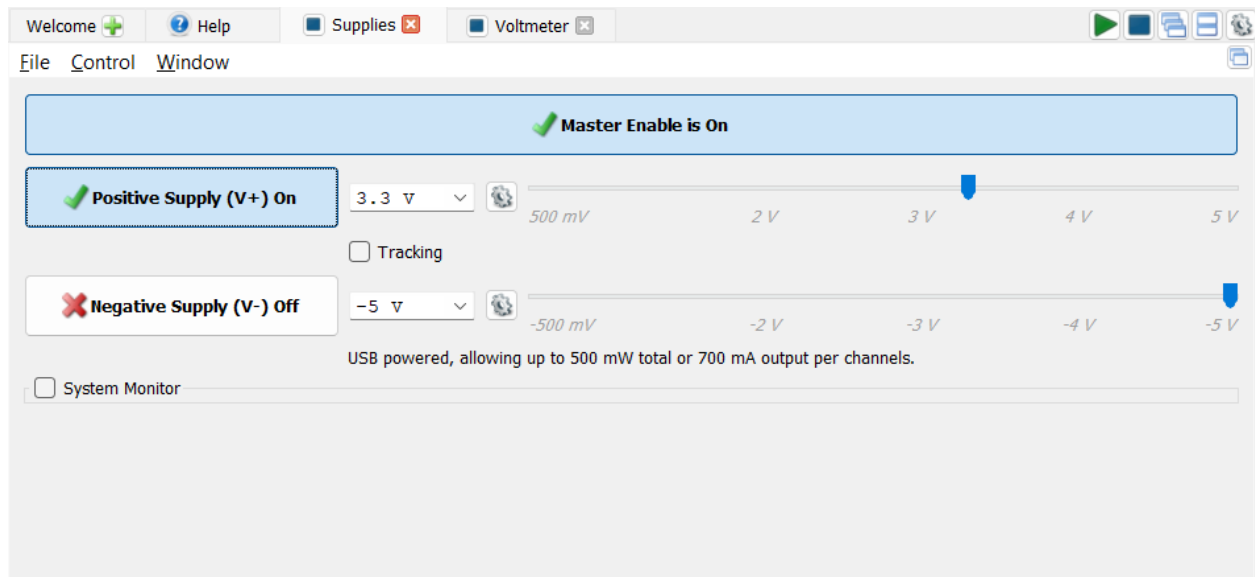
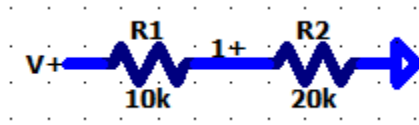
	Channel 1	Channel 2
DC	-3.306 V	2 mV
True RMS	3.306 V	2 mV
AC RMS	2 mV	0 V

3.3V:

Red: V+, 3.3V

Black: Gnd

Blue: 1+, Channel 1



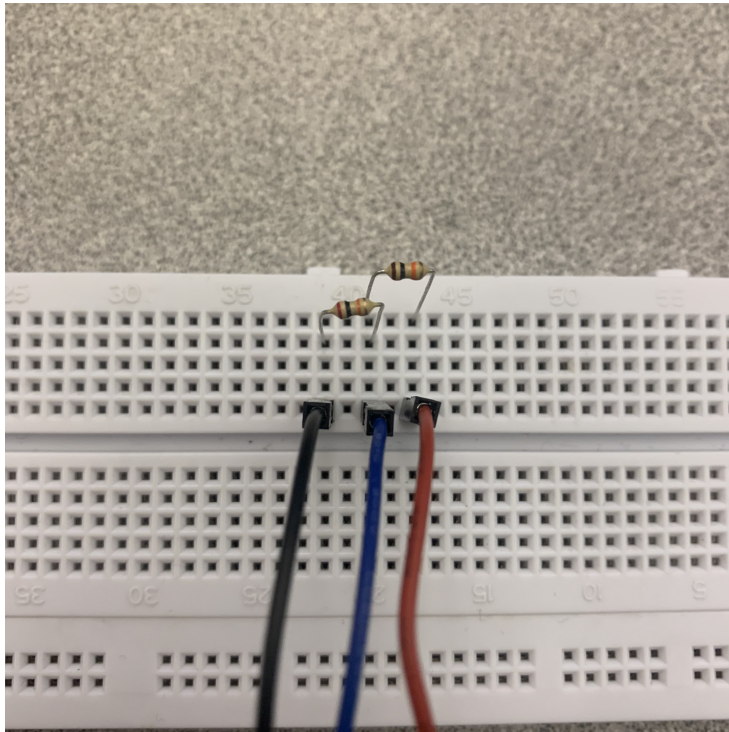
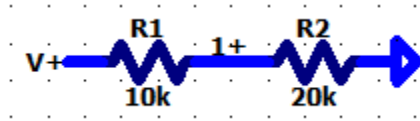
	Channel 1	Channel 2
DC	2.2 V	2 mV
True RMS	2.2 V	2 mV
AC RMS	2 mV	0 V

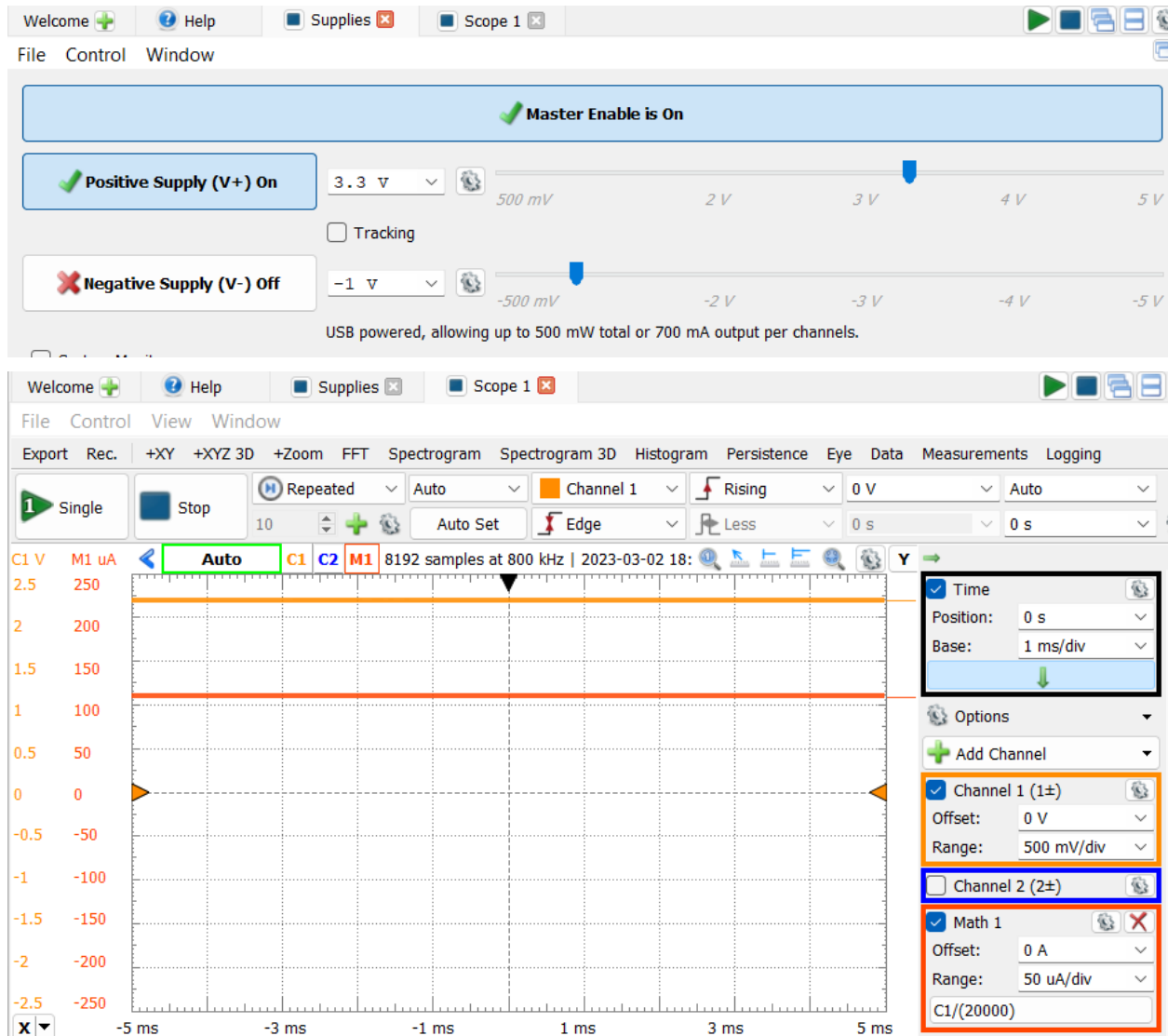
Measure DC current through a resistor: can't be done directly with AD2, used $I=V/R$ with a math channel to calculate it in real time

Red: V+, 3.3V

Black: Gnd

Blue: 1+, Channel 1





I had points taken off for the background not being white. Assuming my vision is correct, that is white.

I also had points taken off for not increasing the line width, but it's also already at the maximum setting.

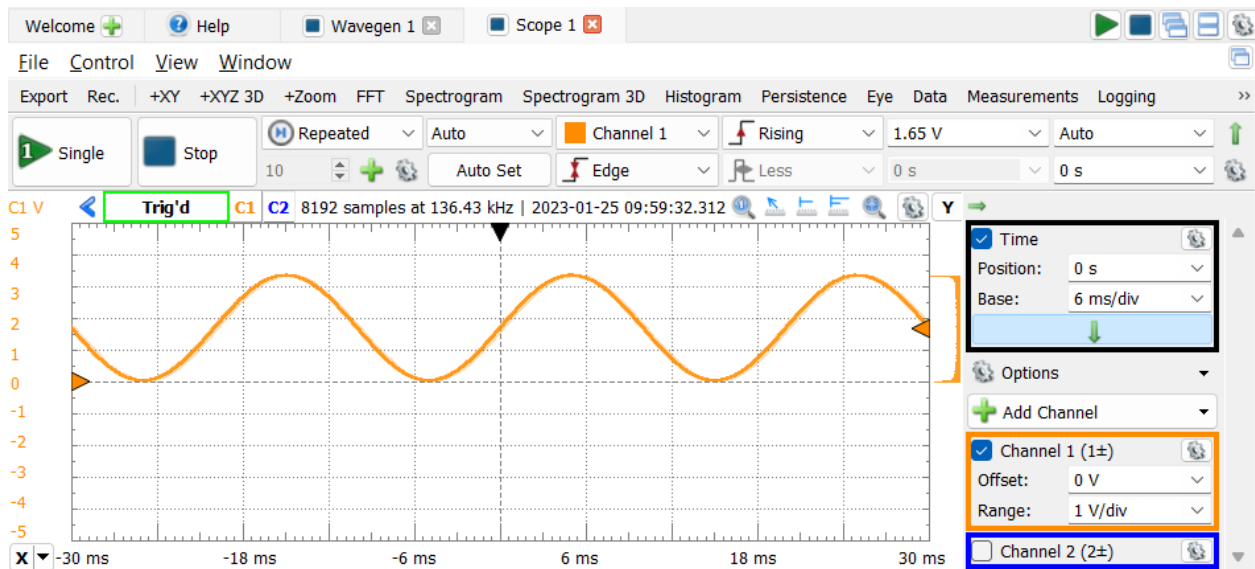
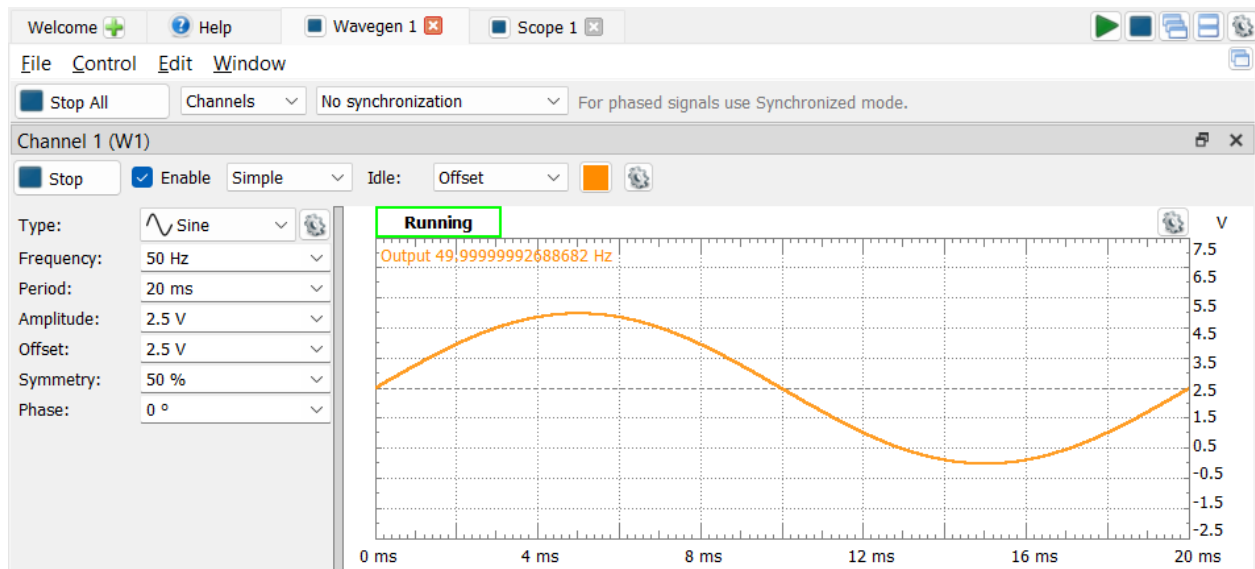
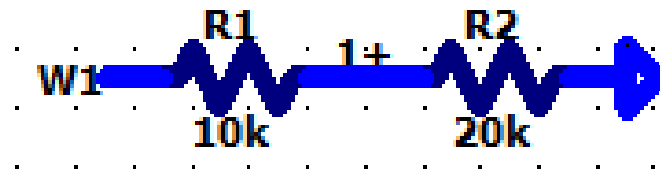
Measure voltage across one resistor with a sinusoid source:

5 to 0:

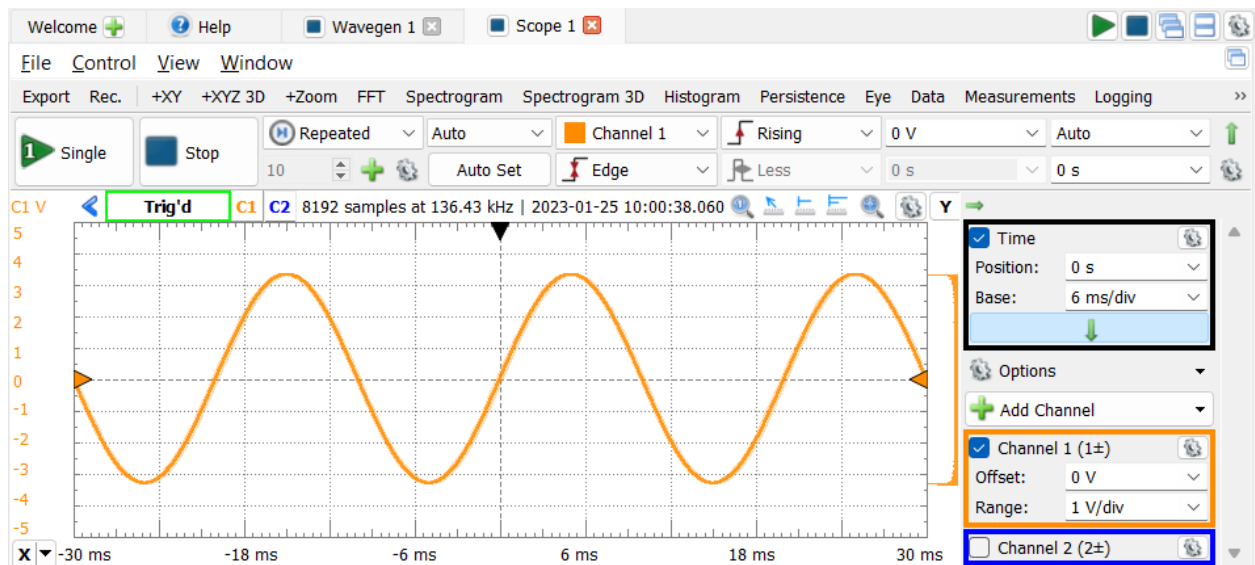
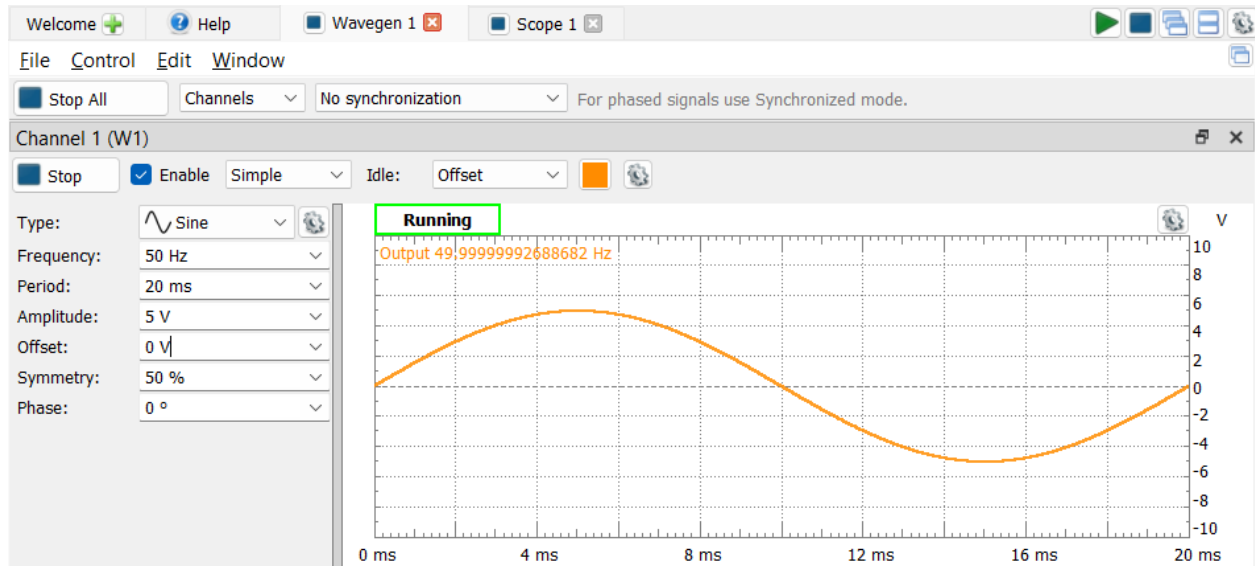
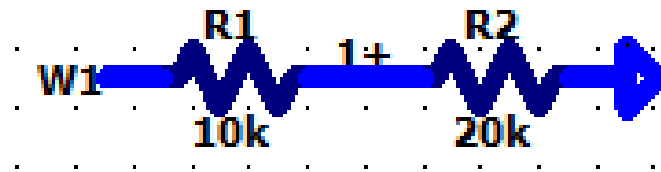
Red: W1

Black: Gnd

Blue: 1+, Channel 1

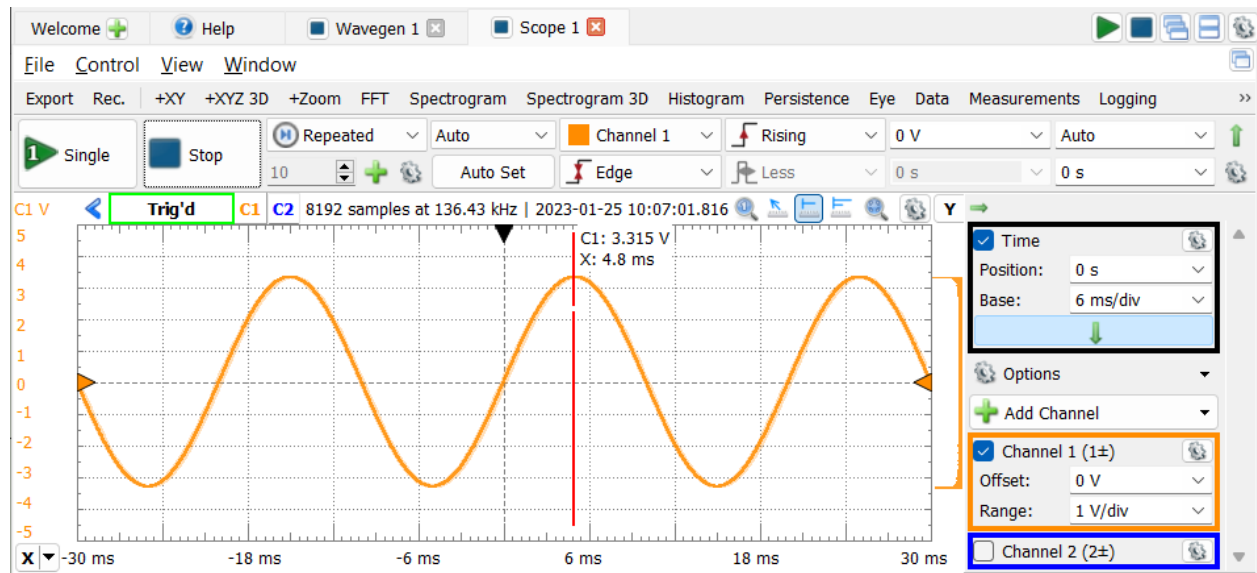


5 to -5:
 Red: W1
 Black: Gnd
 Blue: 1+, Channel 1

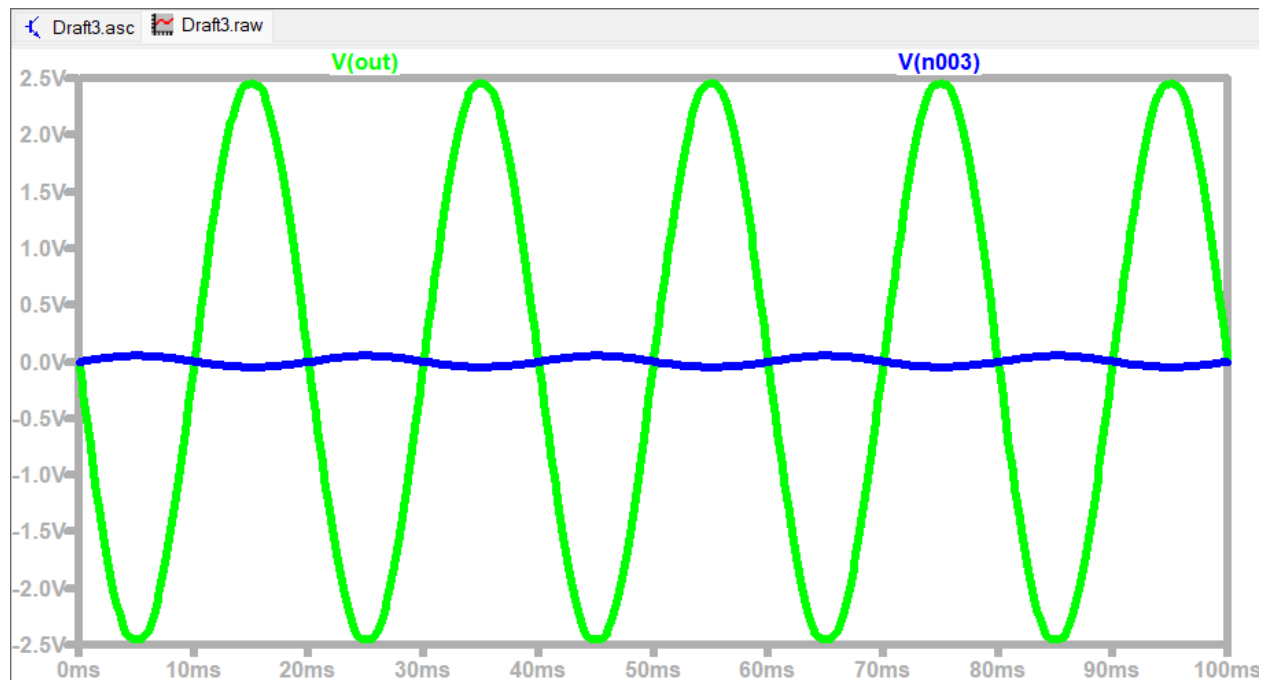
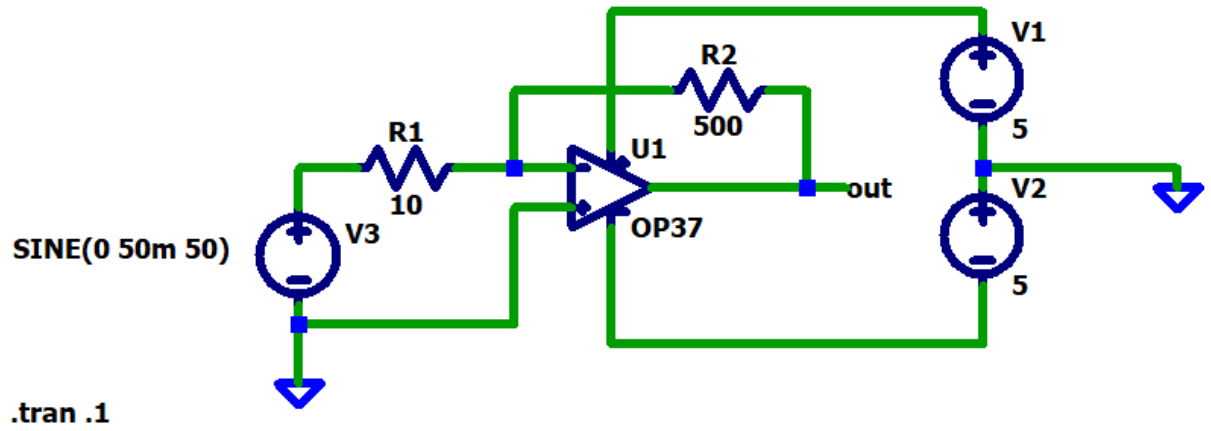


The analog discovery 2 can do both.

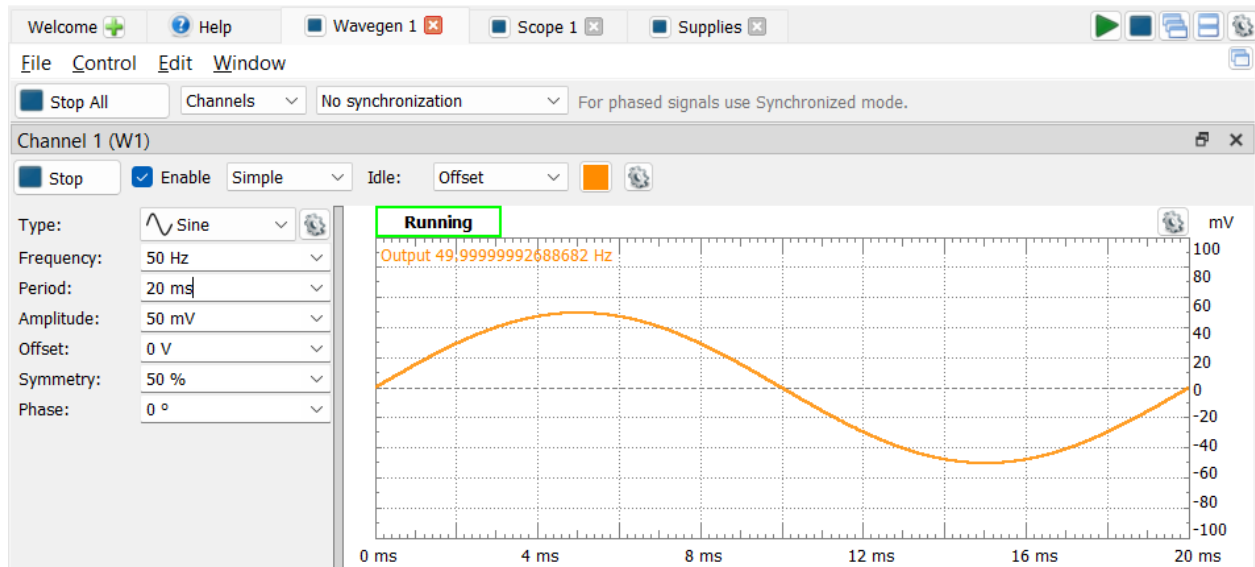
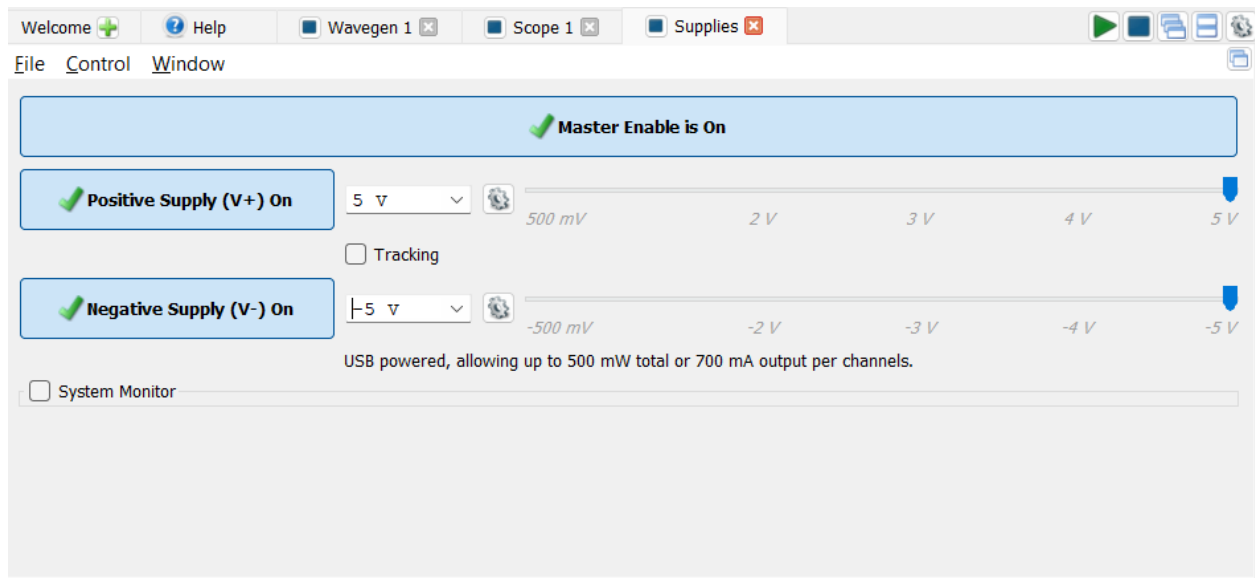
Use cursor function:



Op amp: Inverting amplifier with a gain of 50
LT spice circuit and simulation



Waveforms setup



output

